LUIS ESTEBAN SERRANO BERMÚDEZ

Embedded Systems Engineer | Firmware | Hardware/Software Integration | Embedded Linux

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PROFESSIONAL PROFILE

Computer Engineer specializing in Embedded Software, Firmware, and Embedded Linux, with experience in Python, C/C++, and hardware/software integration. Experienced in biomedical systems, microcontroller firmware development, and peripheral integration using I2C, SPI, UART, GPIO, PWM, and ADC. Skilled in data communication, hardware debugging, and technical documentation.

PROFESSIONAL EXPERIENCE

Arroba Ingeniería - Mexico City 2024 – Present
Firmware Engineer

- Modernized the technology stack of a dual-purpose biomedical device (Incubator/Warmer), migrating the control system and graphical interface to Embedded Linux.
- Developed control logic in Python and the graphical interface using HTML, CSS, and JavaScript.
- Performed hardware/software integration using I2C, SPI, and UART, managing peripherals through GPIO, PWM, ADC, and Device Tree.
- Managed the development lifecycle using Git/GitHub, technical documentation, and continuous delivery of development progress.

Biossmann - Mexico City 2021 - 2024
Firmware Developer

- Analyzed and optimized control software for medical systems and smart operating rooms, integrating lighting devices and operating tables.
- Worked with electronic design teams to diagnose and resolve hardware/software interoperability issues.
- Re-engineered the vital signs monitoring system by integrating video SDKs, Web technologies, and real-time data transmission using HL7.
- Formalized technology transfer through technical documentation, facilitating implementation, scaling, and support by IT and hospital service teams.

ACHIEVEMENTS

- Reverse-engineered and analyzed the code of two critical applications used in hospital systems: one for smart operating room control and another for vital signs monitoring.
- The analyzed applications are currently in use across multiple private hospitals and are undergoing approval for implementation in institutions such as IMSS and ISSSTE.

EDUCATION

Faculty of Engineering - National Autonomous University of Mexico (UNAM)

- Computer Engineering 2014-2020

COURSES

- AUTOSAR Architecture (Learn from Scratch with Demo) — Udemy, 2025
- Software Testing & Debugging — Udemy, 2021
- Git for Developers Using GitHub — Coursera, 2020
- Qt/PySide — Udemy, 2024
- MATLAB/Simulink & DSP — IEEE UNAM Student Branch, 2020

LANGUAGES

- English: B1
- Spanish: Native